$$\dfrac{c = b}{\vdash \text{type\_of }(c) = b}\,\text{Base}_{\text{type\_of}} \qquad \dfrac{w = A}{\vdash \text{type\_of}(w) = A}\,\underset{\text{type\_of}}{\rightarrow}$$

$$\dfrac{\vdash \text{type\_of }(w_1) = t_1 \qquad \vdash \text{type\_of }(w_2) = t_2}{\vdash \text{type\_of }(w_1, w_2) = (t_1 \times t_2)}\,\times_{\text{type\_of}}$$

RULES FOR w>>t :

$$\dfrac{c \in [\![b]\!]}{\vdash c \gg b}\,\text{Base}_w \qquad \dfrac{w \leqslant A}{\vdash w \gg A}\,\underset{w}{\rightarrow} \qquad \dfrac{\vdash w_1 \gg t_1 \qquad \vdash w_2 \gg t_2}{\vdash (w_1, w_2) \gg (t_1 \times t_2)}\,\times_w$$

$$\dfrac{\vdash w \gg t' \qquad t' \leqslant t}{\vdash w \gg t}\,\text{Sous-type}_w$$

RULES FOR $t \rightsquigarrow w$:

$$\dfrac{w = c \leqslant b}{\Delta \vdash_s b \rightsquigarrow w}\,\text{Base}_t \qquad \dfrac{w \leqslant A}{\Delta \vdash_s A \rightsquigarrow w}\,\underset{t}{\rightarrow} \qquad \dfrac{\Delta \vdash_m t_1 \rightsquigarrow w_1 \qquad \Delta \vdash_m t_2 \rightsquigarrow w_2}{\Delta \vdash_s t_1 \times t_2 \rightsquigarrow (w_1, w_2)}\,\times_t$$

$$\dfrac{\Delta \vdash_s t_1 \rightsquigarrow w}{\Delta \vdash_s t_1 \vee t_2 \rightsquigarrow w}\,\vee_{1_t} \qquad \dfrac{\Delta \vdash_s t_2 \rightsquigarrow w}{\Delta \vdash_s t_1 \vee t_2 \rightsquigarrow w}\,\vee_{2_t} \qquad \dfrac{\Delta \cup \{t\} \vdash_s t \rightsquigarrow w}{\Delta \vdash_m t \rightsquigarrow w}\,t \notin \Delta$$

Tree for $t = (\text{Int}, (\text{Int} \to \text{Bool}) \vee \text{Nil})$:

$$\dfrac{\dfrac{\dfrac{w_1 = 42 \leqslant \text{Int}}{\Delta = \{\text{Int}\} \vdash_s \text{Int} \rightsquigarrow w_1}\,\text{Base}_t}{\vdash_m \text{Int} \rightsquigarrow w_1}\,\text{Int} \notin \Delta \qquad \dfrac{\dfrac{\dfrac{w_2 = \text{Int} \to \mathbb{0} \leqslant \text{Int} \to \text{Bool}}{\Delta \vdash_s \text{Int} \to \text{Bool} \rightsquigarrow w_2 = \text{Int} \to \mathbb{0}}\,\underset{t}{\rightarrow}}{\Delta = \{(\text{Int} \to \text{Bool}) \vee \text{Nil}))\} \vdash_s (\text{Int} \to \text{Bool}) \vee \text{Nil}) \rightsquigarrow w_2}\,\vee_{1_t}}{\vdash_m t = (\text{Int} \to \text{Bool}) \vee \text{Nil}) \rightsquigarrow w_2}\,t \notin \Delta}{\vdash_s (\text{Int}, (\text{Int} \to \text{Bool}) \vee \text{Nil}) \rightsquigarrow (w_1, w_2)}\,\times_t$$

Tree for (42, Int -> O) : (Int, (Int -> Bool) or Nil)

$$\dfrac{\dfrac{42 \in [\![\text{Int}]\!]}{\vdash 42 \gg \text{Int}}\,\text{Base}_w \qquad \dfrac{\dfrac{\text{Int} \to \mathbb{0} \leqslant (\text{Int} \to \text{Bool})}{\vdash \text{Int} \to \mathbb{0} \gg (\text{Int} \to \text{Bool})}\,\underset{w}{\rightarrow} \qquad (\text{Int} \to \text{Bool}) \leqslant (\text{Int} \to \text{Bool}) \vee \text{Nil}}{\vdash \text{Int} \to \mathbb{0} \gg (\text{Int} \to \text{Bool}) \vee \text{Nil}}\,\text{Sous-type}_w}{\vdash (42, \text{Int} \to \mathbb{0}) \gg (\text{Int}, (\text{Int} \to \text{Bool}) \vee \text{Nil})}\,\times_w$$

Exemple w2 :

$$\dfrac{\dfrac{3 \in [\![\text{Int}]\!]}{\vdash 3 \gg \text{Int}}\,\text{Base}_w \qquad \dfrac{\dfrac{\text{True} \in [\![\text{Bool}]\!]}{\vdash \text{True} \gg \text{Bool}}\,\text{Base}_w \qquad \dfrac{\text{Int} \to \text{Int} \leqslant \mathbb{0} \to \mathbb{1}}{\vdash \text{Int} \to \text{Int} \gg \mathbb{0} \to \mathbb{1}}\,\underset{w}{\rightarrow}}{\vdash (\text{True}, \text{Int} \to \text{Int}) \gg (\text{Bool}, \mathbb{0} \to \mathbb{1})}\,\times_w}{\vdash (3, (\text{True}, \text{Int} \to \text{Int})) \gg (\text{Int}, (\text{Bool}, \mathbb{0} \to \mathbb{1}))}\,\times_w$$

Règles pour les var de type :

$$\dfrac{\text{Vars}(t) = \emptyset}{\vdash \text{polyw}(t) = \emptyset}\,\text{Base}_{\text{polyw}} \qquad \dfrac{\forall \sigma, t\sigma \not\leqslant \mathbb{0}}{\vdash \text{polyw}(t) = \bigcup_{\alpha \in \text{Vars}(t)} \{\alpha \mapsto \mathbb{0}\}}\,\text{Tally}_{\text{polyw}}$$

$$\dfrac{\sigma' = \{\alpha \mapsto \neg\alpha\sigma''\sigma_{\text{fix}}\} \qquad \vdash \text{polyw}(t\sigma') = \sigma}{\vdash \text{polyw}(t) = \sigma' \cup \sigma}\,\text{Gen}_{\text{polyw}}$$

$$\frac{\vdash \mathrm{Polyw}(\alpha) = \sigma \qquad \Delta \vdash {}_m \alpha\sigma \rightsquigarrow w}{\Delta \vdash {}_s \alpha \rightsquigarrow (\sigma, w)}\,\mathrm{var}_t$$

$$\frac{\mathrm{Vars}(\alpha\sigma) = \emptyset \qquad \alpha\sigma \not\leq \mathbb{0} \qquad \vdash \mathrm{type\_of}(\sigma, w) = \alpha\sigma}{\vdash \mathrm{type\_of}(\sigma, w) = \alpha}\,\mathrm{var}_{\mathrm{type\_of}}$$

$$\frac{\mathrm{Vars}(\alpha\sigma) = \emptyset \qquad \alpha\sigma \not\leq \mathbb{0} \qquad \vdash w \gg \alpha\sigma}{\vdash (\sigma, w) \gg \alpha}\,\mathrm{var}_w$$

exemple polyw1 : $t = \alpha \wedge \beta$

$$\frac{\sigma' = \{\alpha \mapsto \mathbb{1}\} \qquad \dfrac{\sigma'' = \{\beta \mapsto \mathbb{1}\} \qquad \dfrac{\mathrm{Vars}(\mathbb{1}) = \emptyset}{\vdash \mathrm{polyw}((\mathbb{1} \wedge \beta)\sigma'' = \mathbb{1} \wedge \mathbb{1} = \mathbb{1}) = \{\}}\,\mathrm{Base}_{\mathrm{polyw}}}{\vdash \mathrm{polyw}((\alpha \wedge \beta)\sigma' = \mathbb{1} \wedge \beta) = \sigma'' \cup \{\}}\,\mathrm{Gen}_{\mathrm{polyw}}}{\vdash \mathrm{polyw}(\alpha \wedge \beta) = \sigma' \cup \sigma'' \cup \{\} = \{\alpha \mapsto \mathbb{1}\}}\,\mathrm{Gen}_{\mathrm{polyw}}$$

$$
\dfrac{\dfrac{\dfrac{\sigma' = \{\alpha \mapsto \mathbb{1}\} \qquad \dfrac{\mathrm{Vars}(t) = \emptyset}{\vdash \mathrm{polyw}(\alpha\sigma' = \mathbb{1}) = \{\}}\,\mathrm{Base}_{\mathrm{polyw}}}{\vdash \mathrm{polyw}(\alpha) = \sigma' \cup \{\} = \{\alpha \mapsto \mathbb{1}\}}\,\mathrm{Gen}_{\mathrm{polyw}} \qquad \dfrac{\dfrac{42 \leqslant \mathbb{1}}{\{\mathbb{1}\} \vdash_s \mathbb{1} \rightsquigarrow 42}\,\mathrm{Base}_t}{\vdash_m \alpha\{\alpha \mapsto \mathbb{1}\} = \mathbb{1} \rightsquigarrow 42}\, t \notin \Delta}{\vdash_s \alpha \rightsquigarrow (\{\alpha \mapsto \mathbb{1}\}, 42)}\,\mathrm{var}_t}{\vdash_s \alpha \vee \beta \rightsquigarrow (\{\alpha \mapsto \mathbb{1}\}, 42)}\,\vee_t
$$